

(a) The Spatial-Frequency Asymmetry Paradox

Spatial RGB Backbones (ConvNeXt, Xception):

- [x] Memorize biometric actor shortcuts (A -> B)
 - [x] Overfit background scenes; fail on zero-shot fakes
 - [!] Robust to blur, but blind to subtle sensor anomalies
- Result: High in-domain AUC (>99%), collapse on unseen splits*

Standalone Frequency Detectors (SRM, 2D FFT):

- [x] Extreme sensitivity to transmission channel noise
 - [x] Low-pass blur destroys high-pass PRNU residuals
- [!] Polarity Inversion: pristine compressed images**
misclassified as fakes (ROC AUC drops below 0.50)

(b) Physical Acquisition vs. Generative Artifacts

Real Camera Silicon Fingerprint (PRNU):

- * Multiplicative silicon wafer imperfections: K_{PRNU}
 - * High-pass residual power: $\sigma_{real} = 0.28$ (continuous)
 - * Generative decoder PRNU annihilation: $\sigma_{fake} = 0.04$
- Result: Indelible micro-noise contrast in pristine frames*

Transposed Convolution Aliasing & Blending:

- * Transposed conv upsampling emits periodic Dirac spikes
- * +18 dB high-frequency spectral elevation over 1/f decay
- * Poisson boundary blending produces sharp C1 gradient
step discontinuities along facial perimeter ($sf = 1.50x$)

(c) Proposed Solution: SNR-Adaptive Mediation

Dynamic Noise Attenuation Factor (gamma):

- $\gamma = \text{clip}((P_{noise} - \epsilon_0) / (\epsilon_1 - \epsilon_0), 0, 1)$**
- * High Noise Power (Clean Stream): $\gamma \rightarrow 1.0$
Full spectral power leverages PRNU & Dirac spikes
 - * Low Noise Power (Compressed / Blurred): $\gamma \rightarrow 0.0$

Effective Complementary Gated Fusion:

- $\mathbf{g}_{eff} = [\mathbf{g}_s, \mathbf{g}_f * \gamma]^T \Rightarrow \mathbf{f}_{fused} \text{ in } \mathbb{R}^{1024}$**
- [+] Prevents spectral polarity inversion under compression
 - [+] Preserves 0.6722 ROC AUC even under extreme blur ($\sigma=4.0$)
- [+] Reaches 0.8656 Frame / 0.8994 Video AUC on Disjoint Test**